

## PROFILE

Full-stack Software Engineer with 6+ years of experience in building complex, frictionless web applications. Known for tackling technical challenges with ease and rapidly mastering unfamiliar technologies in dynamic Agile environments. Proven leader with experience in managing teams and driving projects from concept to delivery in early- to mid-stage startup environments.

## EXPERIENCE

### Worlds Enterprises Inc.

#### UI Team Lead

Apr. 2025 – Present

- Led the development of an AI chat interface written in TypeScript and React, which communicates with an LLM (Gemma) to find and describe objects that have been previously detected by Worlds' computer vision models.
- Led the development of AIMS, a monitoring interface for the US Air Force using React and TypeScript, ensuring real-time security compliance on military bases.
- Developed Worlds NQ using React and TypeScript, enabling data scientists to efficiently tag training data for object detection models. Decreased model deployment time by about 75%.
- Led a team of two engineers, translating product goals into requirements, features, and tasks. Prioritized and assigned work while mentoring team members and conducting code reviews to ensure delivery.
- Collaborated with leadership and business stakeholders to align project planning with strategic objectives and drive execution.

#### Software Engineer

Feb. 2019 – Apr. 2025

- Led the design and development of the Worlds API and its documentation: A GraphQL API written in Java Spring Boot that exposes data generated by the Worlds platform to the public. Enables forward-deployed and client engineering teams to self-serve integrations, reducing the internal engineering team's bespoke solution workload by about 90%.
- Designed and developed Worlds Navigator in TypeScript, React, and Babylon.js, which simulates clients' site activity and events in real-time within their digital twins.
- Proposed, designed, and developed Worlds Builder using TypeScript, React and Babylon.js on the front-end and Java and Spring Boot on the back-end. Allows internal support teams and clients to create and maintain digital twins. Resulted in a 50% decrease in client onboarding time.
- Developed Worlds Calibration, a Python microservice that enables the real-time simulation of clients' digital twins by attaching global positioning data to objects detected by computer vision models. The underlying logic has been incorporated into multiple awarded patents. (US-11928864, US-12106569, US-11887374, US-11776275)
- Designed and developed bespoke solutions in Java and Python that performed clients' business logic, automated their procedures, and interacted with their physical and technical infrastructure.

## EDUCATION

### Bachelor of Science in Computer Science

*The University of Texas at Dallas*

Aug. 2015 – May 2019

## SKILLS

#### Languages

- TypeScript
- Java
- Python
- HTML/CSS
- C#

#### Frameworks

- React
- Spring Boot
- Node.js

#### Databases

- PostgreSQL

#### Infrastructure

- Kubernetes
- Docker
- RabbitMQ
- Redis
- AWS (S3, ECS, ECR)

#### Tools

- Git/GitHub
- Cursor
- Jira
- Asana
- Unity